Microstructural Aspects of Adiabatic Shear Failure in Annealed Ti6Al4V

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This work reports a comprehensive examination of the microstructural evolution in Ti6Al4V subjected to high-strain-rate deformation. The sequence of microstructural rearrangements leading to adiabatic shear banding is presented. A detailed microstructural comparison between two types of specimens, one that failed by adiabatic shear and the other that was strained to half its failure strain, is carried out. The main observation is that for this material, the microstructure of the two types of specimens is qualitatively identical, indicating that from approximately half the failure strain until adiabatic shear failure, no additional micromechanism is observed to develop and operate. Overall, the microstructure undergoes a significant refinement with the increasing strain until the formation of dynamically recrystallized grains. It is therefore suggested that the evolution of the volume fraction of recrystallized grains should be characterized from its early onset until final failure by adiabatic shear banding.

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I. INTRODUCTION

ADIABATIC shear failure is a dynamic failure mechanism that occurs in many ductile materials subjected to impact loading. The characteristic feature of this mechanism is the development of a narrow band or plane (adiabatic shear band (ASB)) in which very large local strains and high temperatures prevail, resulting in uncontrolled failure.[1,2]

Adiabatic shear banding has been traditionally linked to thermally induced strain softening, leading to a loss of stability of the plastic deformation.[3–5] While a profusion of reports on the microstructural aspects of ASBs can be found (Reference 6, for example, for a review), coordinated mechanical-microstructural experiments on this subject are still very scarce. Recently, Rittel *et al.*[7] proposed to reanalyze the onset of adiabatic shear banding in terms of the dynamically stored energy of cold work in the material. Being an energetic concept, the latter connects naturally to microstructural rearrangements occurring in the material during the dynamic process. In parallel, one finds a very large number of reports of dynamic recrystallization (DRX) observed in the broken ASB in numerous metals.[6,8–16] This observation is again justified in terms of the high temperature inside the band. A contradictory observation was reported by Rittel *et al.*[17] for annealed Ti6AL4V, by which the authors observed dynamically recrystallized grains in specimens that had not failed and were only loaded to roughly half their failure strain by ASB. This striking observation showed that contrary to common belief, DRX actually precedes ASB rather than being its consequence. A related suggestion was that DRX is in fact driven by the dynamically stored energy of cold work, which acts as the thermodynamic driving force of the process. Yet, the following two central issues remain open.

(1) What is the sequence of microstructural rearrangements leading to ASB?
(2) What are the kinetics of the DRX growth until a full ASB is formed?

The first issue is principally addressed in this article. Specifically, the microstructure of the specimens used in Reference 17 is analyzed in great detail in order to pinpoint the microstructural resemblances and the differences between the failed and deformed specimens. The idea is that because the locus for ASB formation is identified *a priori* as the fillet of the gage section (stress concentration), the microstructure can be systematically characterized as a function of the distance from the fillet, revealing the various degrees to which it evolves. Based on this approach, both failed and partially deformed (interrupted) specimens can be characterized and compared, because the failure locus is well defined, as discussed in the sequel, whether as a potential or an actual fracture plane.

Extensive research has been devoted to the microstructural evolution in shear localization areas. A recent extensive review by Xu *et al.*[6] summarized the microstructural evolution related to adiabatic shear localization for strain rates of 10^3 to 10^4 s^{-1} in various metals, including Ti and its alloys. The primary metallurgical process occurring during high-strain-rate deformation is believed to be dynamic recovery in the form of dislocation cell formation, similar to the microstructural evolution of metals subjected to severe plastic deformation.[18] Evolution here is to be understood as strain

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related rather than related to the usual temporal implication. As the ASB is approached, the elongated dislocation cells break down and are gradually replaced by small new grains. The microstructural evolution consists, therefore, of a noticeable refinement ending in DRX. A few mechanisms for high-strain-rate DRX have been proposed.[8,11] The sequence of events is mainly described as a process of continuous (rotational) DRX in which while deformation proceeds and the shear band is approached, the degree of misorientation increases. The process ultimately ends in the formation of new nano- or micrograins.[6,9,19–22]

As mentioned before, an abundant literature is available on the microstructural evolution of different alloys with the recurring observation, irrespective of the investigated material, of DRX taking place in the ASB,[11] as evidenced from very small (a few tens of nanometers) grains with a very low dislocation density, indicating a soft undeformed material. The DRX is therefore intimately associated with adiabatic shear failure, with its characteristic large strains and temperatures, although the phenomenon has been shown to be athermal, occurring even at cryogenic temperatures (*e.g.*, Reference 8).

The Ti6Al4V alloy is one of the most attractive engineering alloys due to its low density, high specific strength, resistance to corrosion, and good high-temperature properties. The selected material for this study is annealed Ti6Al4V alloy, which possesses a marked propensity to fail by ASB formation.[1,4,12,23,24] A brief survey of the literature follows in order to summarize the latest developments in the microstructural aspects of the dynamic failure of this and related alloys.

The microstructural aspects of ASB formation in Ti6Al4V were mainly studied following high-strain-rate deformation in the form of ballistic impact and shock loading,[12,15,24–26] in which the typical strain rates were $\dot{\varepsilon} \geq 10^4\,s^{-1}$. Transmission electron microscopy (TEM) observations revealed that at these strain rates, DRX is found within the ASB in the form of grains with diameters a few tens of nanometers in size located in a high-dislocation-density matrix. Lee and Lin[27] investigated this alloy at a lower strain rate ($\dot{\varepsilon} = 10^3\,s^{-1}$) over a large range of temperatures. These authors presented limited microstructural information using TEM, with emphasis on the structure of dislocation cells at various temperatures. The majority of the work performed in the $\dot{\varepsilon} = 10^3$ to $10^4\,s^{-1}$ range concerned α-Ti.[16,28,29] The microstructural evolution following high-strain-rate deformation was found to be similar to that of other metals and alloys that are susceptible to shear band formation, namely the formation of dislocation cells, subgrains, and DRX. Altogether, all the characterized materials bear a common overall microstructural evolution. However, the microstructural evolution in the vicinity of the ASB and within it was not fully reported for Ti6Al4V, to the best of our knowledge.

At this stage, it is important to note that shear band observations are inherently carried out in a high-strain-gradient area. Therefore, by systematically varying the distance of observation from the shear band (fracture plane) itself, various strain levels are in fact interrogated with their corresponding typical microstructure. What has not been carried out is the same kind of systematic observations, comparing now fully fractured and interrupted specimens[17] from a microstructural point of view. Such a comparison provides a qualitative comparative picture of the kinetics of microstructural refinement.

In this work, a comprehensive examination of the microstructural evolution in dynamically deformed and fractured annealed Ti6Al4V is presented, and the sequence of microstructural rearrangements leading to ASB is characterized. This work completes and sheds additional light on the preliminary report of Rittel *et al.*[17] in which the emphasis was mainly put on DRX. The main outcome of the present work is that the microstructure of interrupted specimens, deformed to approximately half of the failure strain in this case, is qualitatively similar to that of fractured specimens to the point that they cannot be distinguished. This point is thoroughly developed in this article.

II. EXPERIMENTAL PROCEDURE

Shear compression specimens (SCSs) were machined from a commercially annealed Ti6Al4V 12.7-mm-diameter rod. The SCS is a specimen in which the kinematics of the deformation enforces a shear dominant situation in the gage section with homogeneous strain and stress fields.[30–32] The specimen shown in Figure 1 consists of a 10-mm-diameter cylinder with a pair of diametrically opposed grooves, making an angle of 45 deg with respect to the longitudinal axis, which delineates the deforming gage section of the specimen. As with any nonsmooth specimen, a mild state of stress concentration develops in the root of the two pairs of fillets (F in Figure 1) that define the gage section,[30] so that ASBs

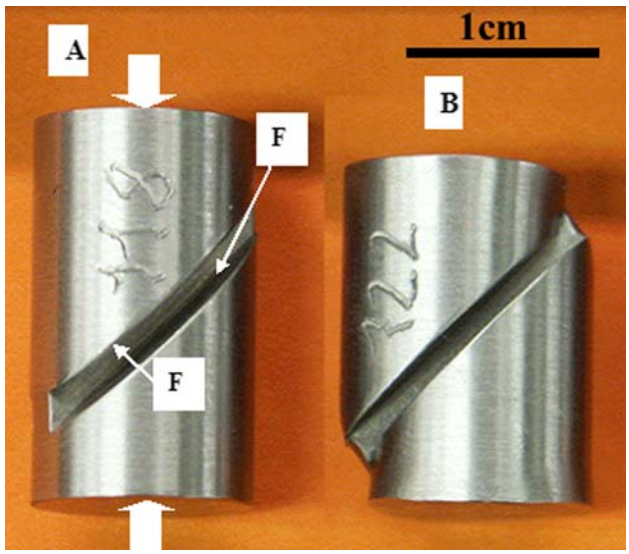


Fig. 1—Typical SCS (*a*) before and (*b*) after deformation. Block arrows show applied load direction and thin arrows point to the stress-concentration fillets (F) where ASBs form. Specimens for TEM were taken from this area.

always initiate in this region of the gage, providing a well-defined initiation site. Dynamic tests were carried out at a typical strain rate of $\dot{\varepsilon} = 3000\,s^{-1}$ using a Kolsky (split Hopkinson pressure bar) apparatus[33] and an SCS specimen with a 1.5-mm gage height, as reported in Reference 17. The dynamic tests consisted of either impacting a specimen until failure (fracture) by ASB formation or bringing it to a controlled level of strain using hardened steel stop rings followed by elastic unloading. The stop ring ensures that the specimens are impacted only once as opposed to being repeatedly pounded by the stress waves trapped in the bars. For these specimens, the normalized strain (with respect to fractured specimens) was approximately 0.45. Additional experiments were carried out quasistatically in which the specimens were deformed to a normalized strain of the approximate same magnitude and unloaded without failure, for comparison purposes.

The TEM specimens were prepared from each type of specimens (interrupted static, interrupted dynamic, and dynamically failed) for a comparative characterization of the microstructure. The specimens were extracted from the fillet region of the specimen, which in the case of fracture contained the shear band. In addition, the initial microstructure of the undeformed material was characterized in order to provide a reference microstructural state. In the deformed samples, the areas set for observation in all specimens included the fracture plane, whether potential or actual, in order to allow for comparative characterization.

For all the unbroken samples, the electron-transparent area was prepared by jet electropolishing (TenuPol-5, Struers Company, Ballerup, Denmark) in a solution of 80 pct methanol and 20 pct HNO_3. For the fractured samples, the transparent area for electrons was prepared by precision ion polishing (Gatan 691 Precision Ion Polishing System (PIPS*)). The two preparation methods yielded representative TEM specimens. However, ion milling was the only feasible way to prepare the transparent area from the fracture plane itself.

*PIPS is a trademark of Gatan, Inc., Pleasanton, CA.

III. RESULTS

A. *Mechanical Behavior*

The results of the mechanical tests can be found in Rittel *et al.*[17] and will only be briefly reminded here. As mentioned earlier, two kinds of tests were performed, namely tests up to fracture and tests interrupted before fracture. Dynamic fracture by ASB formation occurred at a strain level of $\varepsilon_f \approx 0.23$, while the deformed specimens that did not fail (subsequently referred to as interrupted tests) reached a level of $\varepsilon_f \approx 0.12$ (normalized $\varepsilon_n \approx 0.45$) at a strain rate of $\dot{\varepsilon} = 3000\,s^{-1}$. In addition, a static interrupted test was carried out up to $\varepsilon_f \approx 0.17$ (normalized $\varepsilon_n \approx 0.68$) to allow for comparison with its dynamic counterpart. Here, it is important to note that the investigated alloy experiences a very small temperature rise prior to the strain-softening phase.[23] Therefore, for the interrupted test cases, the evolution of heat is even smaller, so that thermal factors in this case can be ruled out.

B. *Microstructural Evolution*

The microstructural evolution of Ti6Al4V was examined using TEM in the vicinity of the shear concentration area, when the field of view was gradually shifted toward the areas of the fracture plane (ASB). The observations were performed over a distance of a few tens of microns for all the samples, and the results will be presented for each sample, starting far away from the shear concentration area then gradually approaching it (including eventual fracture) for the relevant specimens.

1. *Initial microstructure: reference samples*

The initial microstructure of annealed Ti6Al4V alloy consists of α and β phases that are readily identified using selected area diffraction patterns (SADPs). The α-phase grain size is of the order of 5 to 10 μm, while the β-phase grain size is finer, on the order of 1 μm. The grains are essentially free of dislocations, without noticeable twins or martensitic platelets. Figure 2 shows the typical microstructure of the annealed alloy, in which both phases are shown and denoted in the micrograph. A corresponding SADP of the β phase is attached to the micrograph.

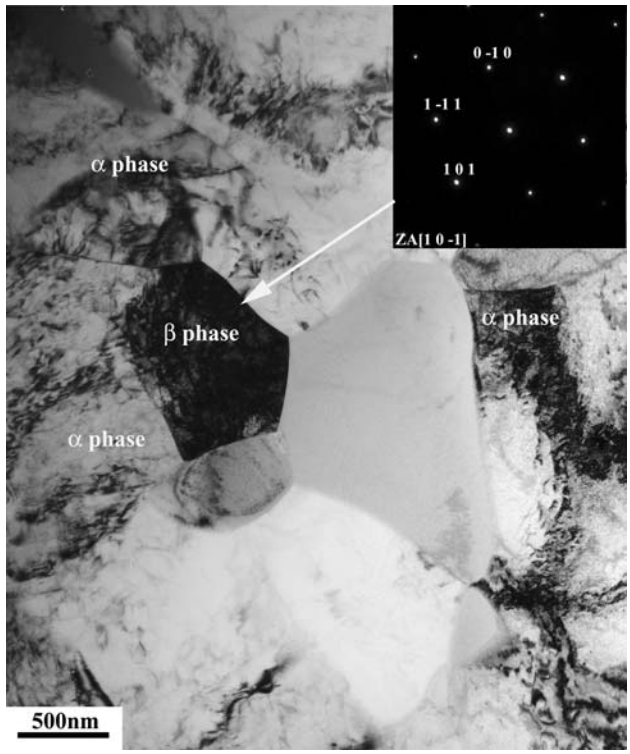


Fig. 2—Reference sample (TEM micrograph): Initial microstructure of annealed Ti6Al4V alloy consists of α and β phases, marked in the micrograph. A β grain is arrowed in the micrograph and a corresponding SADP is attached.

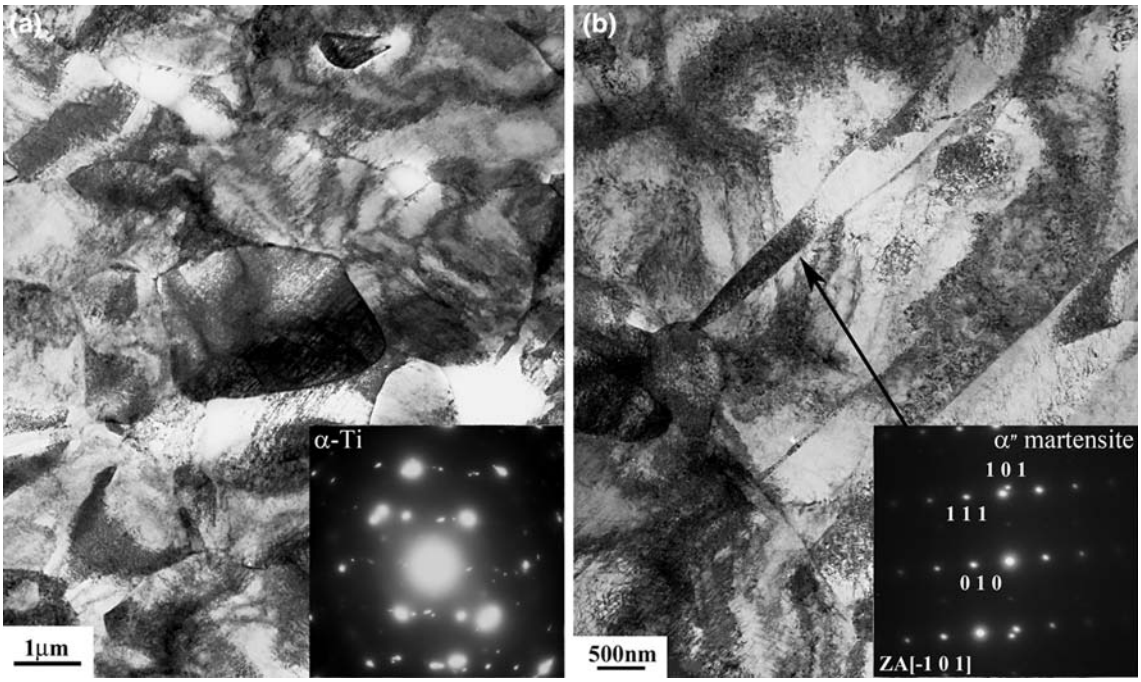


Fig. 3—Interrupted static sample (TEM micrograph): (*a*) typical microstructure consists of dislocation cells within α phase (the corresponding SADP of α-Ti is attached to the micrograph) and (*b*) area showing stress-induced (α″) martensite (arrowed) transformed from the β phase and its corresponding SADP.

2. *Interrupted static specimens*

The microstructure of the interrupted sample in the shear concentration area consists mainly of equiaxed dislocation cells in the α phase, approximately 2 μm in diameter (Figure 3). The interior of the cells contains dislocations, consistent with observations of low-strain quasistatic deformation of high-stacking-fault-energy metals.[18] The cell walls consist of tangled dislocations, which are also typical of low-strain microstructures. The microstructure observed in the interrupted static samples is shown in Figure 3(a). The attached corresponding SADP is identified as α phase, with a small misorientation angle between the dislocation cells.

Martensitic platelets are occasionally observed in the β phase, such as those arrowed in Figure 3(b). These platelets were identified from the corresponding SADP (attached to the micrograph) as α″ (orthorhombic) type, which forms as a result of external stress (*i.e.*, stress-induced martensite) or quenching from a high-temperature β phase.[34,35]

Overall, the microstructure of the interrupted static samples is consistent with that observed at low strains in many high-stacking-fault-energy metals. At these strains, the microstructure consists essentially of dislocation cells that decrease in size and increase in relative misorientation as the strain increases. When moving further toward the shear concentration area, no significant change was observed in the microstructure, even in the fillet edges.

3. *Interrupted dynamic specimens*

Away from the potential fracture plane (~50 μm), the microstructure consists of dislocation cells in the α phase (Figure 4(a)). The cell size is approximately 2 μm, similar to the cell size observed in the interrupted static samples (compare Figures 4(a) and 3(a)). Closer to the potential fracture plane (approximately 10 to 20 μm), the cell walls become sharper, indicating that the dislocation density in the boundaries is higher, and the misorientation angle between adjacent cells is larger than before. The density of martensitic platelets was observed to increase gradually as one gets closer to the shear concentration area (potential fracture plane). In addition, the density of martensitic platelets in the β phase is sensibly higher in the interrupted dynamic samples than in the interrupted static ones (compare Figures 4(a) and 3(a)).

As a general remark, note that two types of martensite were observed in the specimens, irrespective of the strain rate. For example, the martensite in Figure 4(b) is of the α′ (hcp) type, which is also stress induced from the β phase, as was the α″ (orthorhombic (Figure 3(b)).

Closer to the stress concentration fillet (potential fracture plane), the detailed internal structure of the cell walls, which is revealed by tilting the sample, is shown in Figure 5. The cell walls consist of arrays of parallel dislocations, *i.e.*, dislocation networks (square or hexagonal). These dislocation walls introduce a misorientation between adjacent cells, as can be seen in the corresponding SADP, attached to the micrograph manifested by arclike reflections. Such arrays of parallel dislocations were observed neither in the interrupted static nor in the current interrupted dynamic samples in areas that are far away (more than 50 μm) from the fracture plane. The formation of dislocation networks corresponds to the work hardening of the material in the shear localization area.

Closer to the potential facture plane (<10 μm), densely packed dislocations or avalanched dislocation cells are observed (Figure 6). Microstructural features

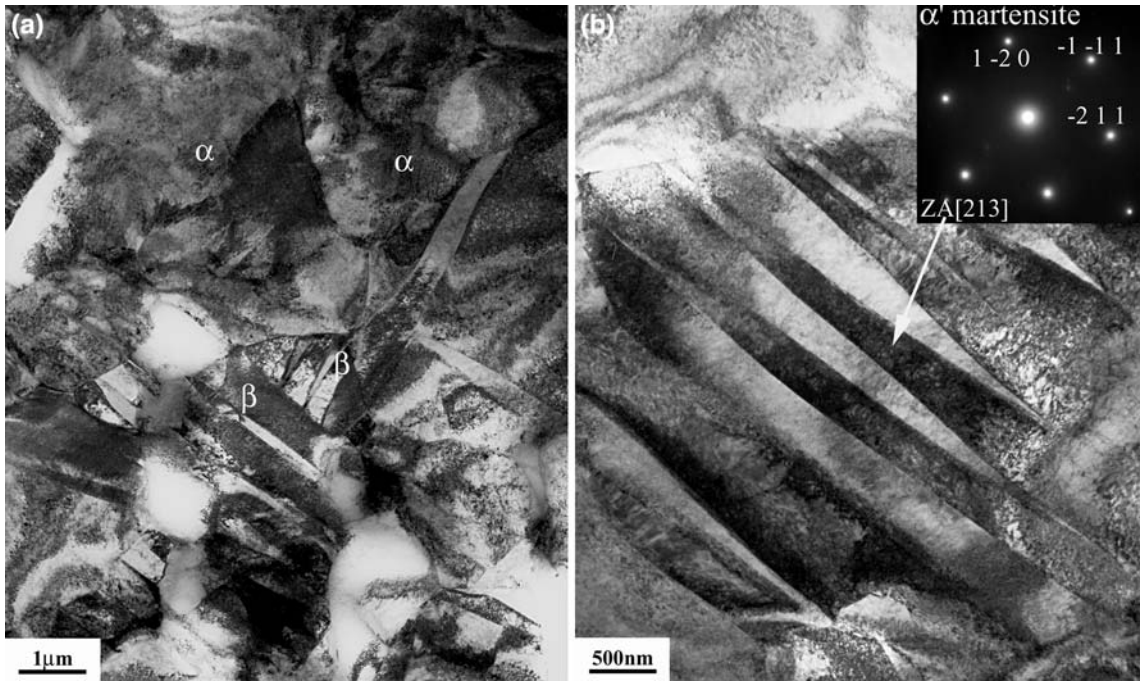


Fig. 4—Interrupted dynamic sample (TEM micrographs) far away from the shear concentration area: (*a*) typical microstructure consists of dislocation cells in α phase and higher density of stress-induced martensite in β phase, both denoted on the micrograph; and (*b*) area showing stress-induced (α') martensite (arrowed) transformed from β phase and its corresponding SADP.

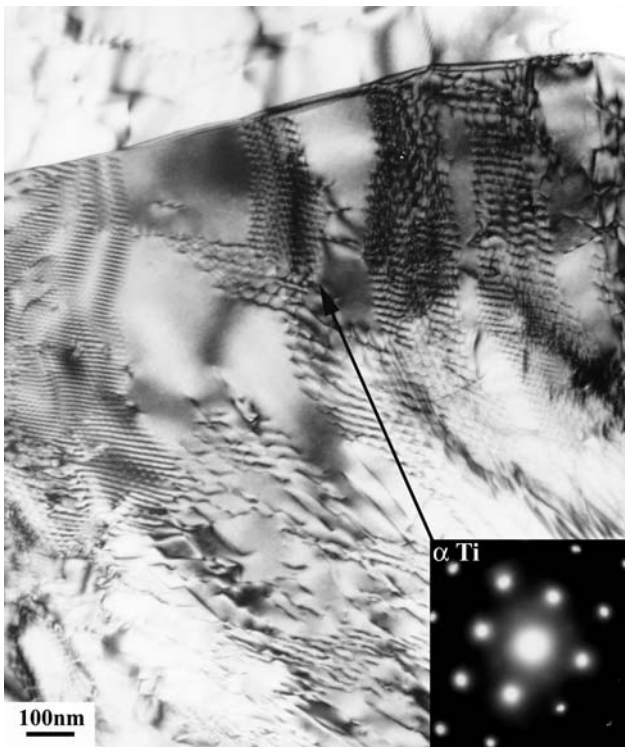


Fig. 5—Interrupted dynamic sample (TEM micrograph) close to the shear concentration area (approximately 10 to 20 μm). Detailed internal structure of dislocation cell walls, consisting of square and hexagonal dislocation networks in α phase. Attached corresponding SADP shows small misorientation angle between adjacent dislocation cells caused by hexagonal network arrowed in the micrograph.

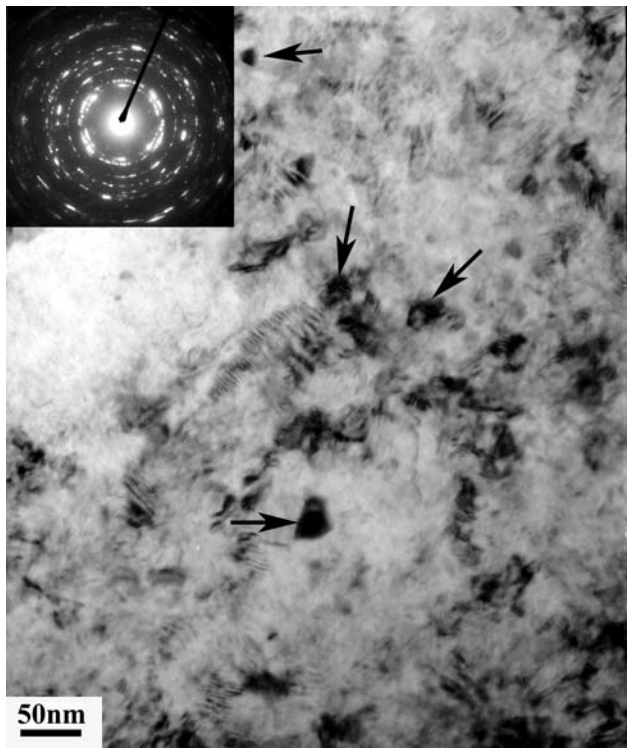


Fig. 6—Interrupted dynamic sample (TEM micrograph) in shear concentration area. Dynamically recrystallized grains formed in highly dislocated area are indicated by arrows. Grains range in size from 10 to 30 nm and have a low dislocation density. Corresponding SADP consists of ring patterns, typical of nanograined polycrystalline materials.

such as well-defined dislocation cells or martensitic platelets are no longer distinguishable. However, SADPs clearly reflect a polycrystalline microstructure, *i.e.*, incomplete rings are observed that indicate the presence of dynamically recrystallized nanograins (for example, the attached SADP in Figure 6). Within the area of avalanched dislocation cells, several nanograins are directly observed (arrowed in Figure 6). These nanograins are a result of DRX and they appear to have a very low dislocation density. Nonetheless, these are rather difficult to discern, because the high density of the surrounding dislocations screens the new dynamically recrystallized nanograins.

The evolution of the microstructure in the interrupted dynamic samples, approaching the potential fracture plane, can thus be summarized as follows: The cell size decreases, the misorientation angle between cells increases, and the cell boundaries become denser, *i.e.*, the cells become subgrains. As one comes even closer to the potential fracture plane, DRX nanograins are observed within an area comprised of avalanched dislocation cells. The corresponding SADP shows incomplete rings consistent with DRX; however, most of the individual nanograins are screened by the dislocations and are thus hard to resolve (compare Figures 6 and 7(c)).

4. *Dynamically failed specimens*

In a similar manner, the microstructural evolution of Ti6Al4V was examined in the shear localization area of fractured samples, where failure due to the formation of an ASB had occurred. The fracture plane was located, as expected, in the fillet of the SCS sample.

The evolution of the microstructure in these samples is overall quite similar to the evolution observed in the interrupted dynamic samples. As the field of view is moved closer to the fracture plane, the microstructure changes gradually from dislocation cells (small misorientation angles between adjacent cells) into subgrains (higher-misorientation angles between subgrains) manifested in the SADP as spotty ring patterns. Figure 7(a) shows the typical microstructure observed in this region, consisting of misoriented subgrains in the α phase and a high density of stress-induced martensite in the β phase. In the core of the fracture plane (shear localization area (Figure 7(b)), a region comprised of avalanched dislocation cells (high-density dislocations) is observed, characterized by incomplete rings in the SADP. Dynamically recrystallized nanograins with a low dislocation density, 10 to 50 nm in diameter, are again observed within the avalanched dislocations region. These DRX nanograins are discernable at higher magnifications, as can be seen in Figure 7(c) (arrowed). Note that a similar evolution was reported by Xue *et al.*[19,20] for 316L-SS.

To summarize this section, it appears that the evolution of the microstructure in the broken specimens is quite similar to that of the interrupted specimens, to an extent to which one cannot tell the difference between them on a qualitative basis.

IV. DISCUSSION

The evolution of the microstructure of commercial Ti6Al4V was examined following high-strain-rate deformation and compared to the static microstructure at a similar strain. The microstructural features observed are presented in a schematic drawing in Figure 8, illustrating microstructural refinement as the shear localization area is approached, keeping in mind that a large strain gradient is present in this region.

The initial microstructure of the Ti6Al4V consists of α and β phases. The microstructure observed in the interrupted static samples is consistent with the microstructure observed following low-strain-level deformation of high-stacking-fault-energy metals.[18] The microstructure consists mainly of dislocation cells with a slight mutual misorientation. Stress-induced martensite is occasionally observed.

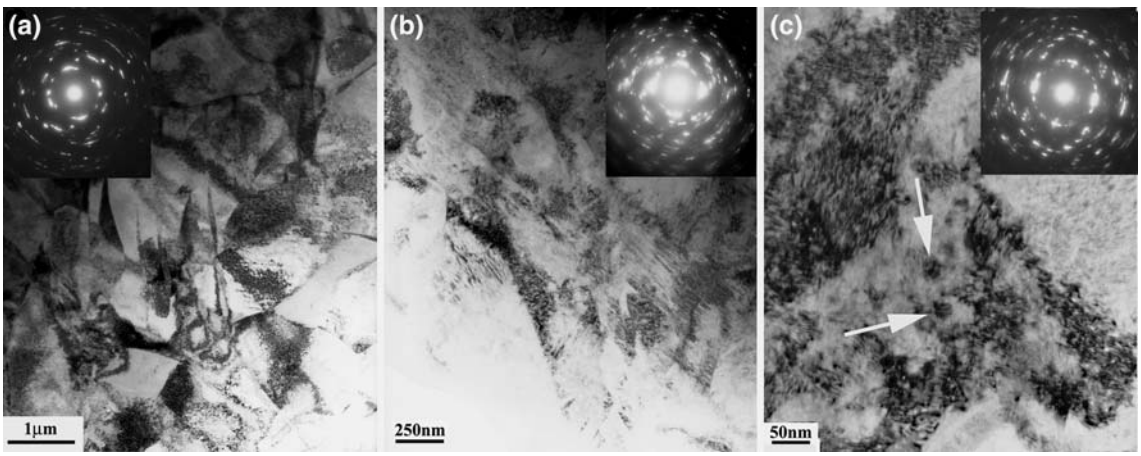


Fig. 7—Dynamically failed specimen (TEM micrographs): (*a*) far from the ASB: dislocation cells and stress-induced martensite (a typical SADP mainly contains misoriented α-phase cells); (*b*) within the ASB: high dislocation density without any distinct morphology (the SADP shows incomplete rings, indicating the presence of very fine recrystallized grains); and (*c*) higher magnification of (b): high-dislocation-density screens the fine grains revealed by the SADP. A few DRX nanograins are arrowed in the micrograph. Note the similarity of morphologies between Figures (6) and 7(c), namely, high dislocation density and the presence of very fine recrystallized grains (reprinted from Ref. 17).

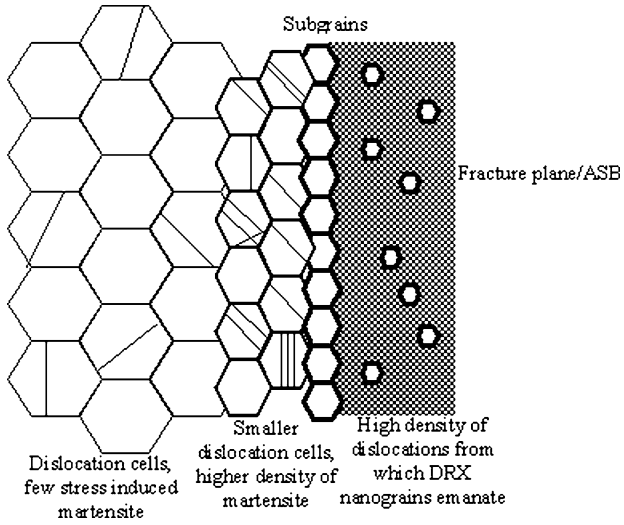


Fig. 8—Schematic representation of microstructural refinement as a function of the distance from ASB/fracture plane. Microstructure evolves from dislocation cells into subgrains and the density of martensite increases with proximity to the fracture plane. In fracture plane, an avalanched, high-dislocation-density area is observed from which DRX nanograins emanate.

Consistent with many previous observations, a pronounced microstructural refinement is observed as the observation area gets closer to the (potential) fracture plane. Far away from this area, the microstructure consists of dislocation cells similar to those observed in static specimens. As the fracture zone is approached (higher strains), the microstructure gets refined by increasing the misorientation angle, resulting in subgrains as observed in similar cases for other materials.[9,11,12,22,28,29] Furthermore, for interrupted tests at the core of the potential fracture plane, an area with a high density of dislocations or avalanched dislocation cells is observed. No microstructural features are distinguishable, even though the corresponding SADPs indicate recrystallization in the form of incomplete rings. Nanograins are discernable within the avalanched dislocation cells area. These nanograins have a very low dislocation density and are a result of DRX in the shear localization area. It seems that the dynamically recrystallized nanograins are screened by the high dislocation density, because the SADP undoubtedly corresponds to many orientations in a small volume, consistent with recrystallization.

The dynamically failed samples exhibit essentially the same microstructural evolution. The refinement of the microstructure as the field of view is moved toward the fracture plane displays the same features, namely, dislocation cells evolve into subgrains with a higher dislocation density within the walls and higher misorientation angles between adjacent volume elements. An avalanched dislocation area is also observed in which DRX nanograins originate.

In other words, no significant microstructural differences are observed between the two dynamically deformed samples, failed and partially deformed. Dynamically recrystallized grains are clearly observed in the two types of specimens, which is surprising for specimens that were not loaded to failure, and barely reached half of its failure strain (Figure 6).[17] Moreover, the recrystallized nanograins seem to have an overall similar shape and dimensions in the two investigated specimens.

As a general remark, one should note the scarcity of work in which the microstructure is tied to the mechanical history of the material, with emphasis on adiabatic shear failure. Here, one can find a few noticeable references.[19,20,36,37] The authors in Reference 36 presented an analysis of DRX in shear bands, with emphasis on the local texture. These authors show an excellent adequation between their analysis and the observed texture of ASBs in steel, as well as Xue *et al.*[19,20] for 316L stainless steel. More recently, DRX in ASBs was successfully modeled using numerical techniques.[37] Here, DRX was represented as a thermally activated phenomenon rather than as a precursor for failure by adiabatic shear banding.

One important result of this study is the microstructural characterization of the gradual evolution that takes place as a function of the local strain, because an inherent gradient exists in the specimen as a result of the strain-concentration effects due to the fillet. In other words, one single specimen was shown to comprise a whole range of increasingly finer microstructures with increasing strain. Consequently, a qualitative description of the gradual refinement was given in Figure 8, which is in good agreement with that reported by Xue *et al.* for 316L stainless steel.[19,20,22]

In addition, this study presents important new evidence related to ASB formation. Essentially, the present work shows that the overall microstructural evolution of annealed Ti6Al4V can be fully observed in dynamically interrupted samples that were only deformed to half their failure strain. These qualitative observations do not allow for a distinction between the two types of specimens, interrupted and failed, because all the microstructural features are already present in the interrupted specimens. The next and only microstructural evolution that characterizes the dynamically failed samples is the very formation of the ASB itself. This formation is believed to be detected in the global mechanical response of the samples (strain softening) and not by any microstructural changes, as shown in this work. This observation suggests that because additional microstructural refinement does not occur past an "incubation phase" during which dynamically recrystallized grains are formed at an early stage, their volume fraction must increase with the ongoing strain until final failure. Consequently, future studies should concentrate on the kinetics of growth of the recrystallized volume fraction of these grains in some quantitative way (question (2) in Section I). Specifically, analyses in the spirit of References 36 and 37 should be carried out, taking into account athermal early generation of DRX and the fact that at this stage, the microstructure is qualitatively "frozen."

Another outcome of this study is that the same recrystallized grains are observed prior to and after shear localization without any apparent change in shape

or dimensions. This seems to contradict the expectation that virgin grains should be highly deformed in the shear band. One could argue, as has been done, that the high temperature and large strains in the shear band promote DRX or could at least anneal the deformed nanograins, thereby reducing their dislocation density. This might happen through diffusional processes as the shear band cools down to room temperature after failure, although this happens in a relatively short time for which diffusional effects might not be significant.

As a related effect, one would naturally consider hardening related to the well-known Hall–Petch effect. In this context, Reference 38 brings valuable insight into the high-strain-rate deformation of nanograined materials. Specifically, for copper, these authors claim that Hall–Petch hardening is expected for all grain sizes at high strain rates as opposed to a potential reverse softening effect at quasistatic strain rates, as a function of the nanograin size. This point remains to be verified experimentally.

Alternatively, one might think of another deformation mechanism other than one involving the slip and motion of dislocations. In this case, in Reference 10, a suggestion is made related to a deformation mechanism that is analogous to superplasticity by which deformation proceeds through grain boundary sliding of the nanograins, with little impact on their initial shape, consistent with the present observations. The few observations of the shape of the nanograins, prior to and after failure, seem therefore to support this hypothesis.

V. CONCLUSIONS

The general evolution of the microstructure of dynamically loaded Ti6Al4V specimens has been characterized as a function of the distance from the fracture plane (ASB).

The main characteristic of this evolution consists of microstructural refinement from dislocation cells with increasing misorientation until the formation of dynamically recrystallized grains. In that respect, this evolution is quite similar to that reported for many other metals, while focused on a specific annealed Ti6Al4V alloy.

An original observation concerns the comparison between specimens deformed to half their failure strain and failed specimens. This comparison shows beyond any doubt that the two microstructures are very similar, with the appearance of dynamically recrystallized grains in the interrupted specimens in spite of their limited deformation.[17] Therefore, long before the ASB has formed, all the characteristic microstructural features are already present in the specimens. Consequently, the ASB must be characterized by a high volume of dynamically recrystallized grains rather than by any new specific feature, a point that was not previously made.

Future work should therefore concentrate on the kinetics of the growth of the recrystallized phase in order to devise a quantitative model leading to the prediction of a fully developed shear band.

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